

LLM Powered Memory Consolidation for Ubiquitous Computing

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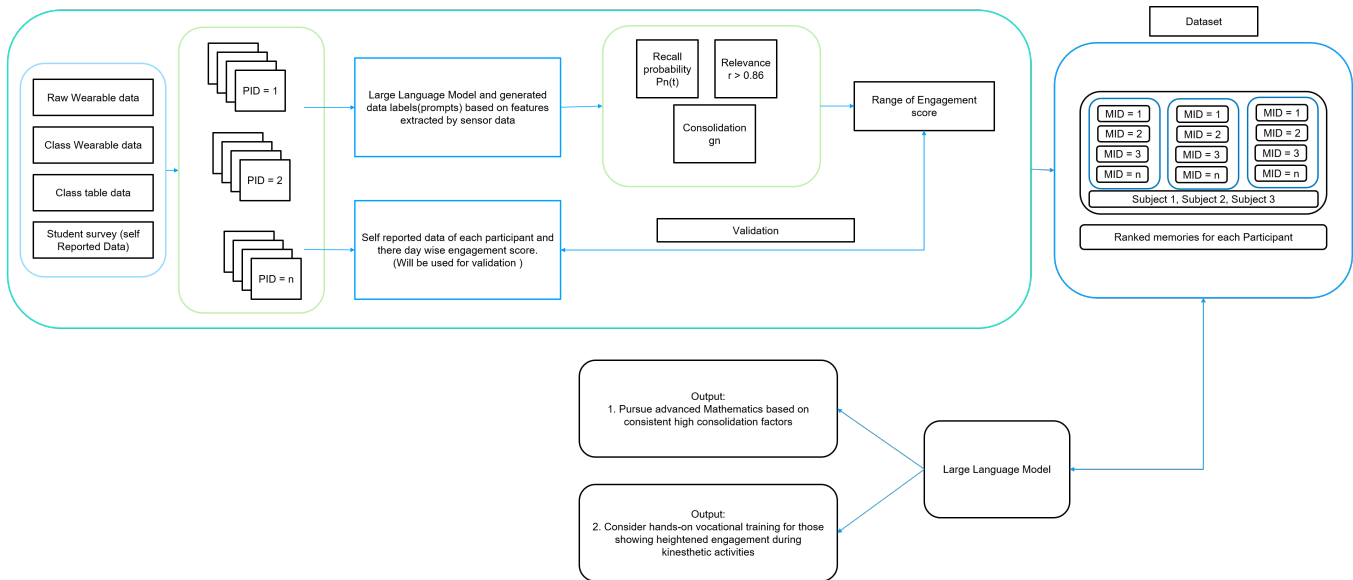


Figure 1: Overview of the multi-modal data processing pipeline for engagement prediction using wearable sensor data and large language models.

Abstract

Large Language Models (LLMs) combined with ubiquitous sensing and computing are increasingly being leveraged to design and architect AI agents for generative and conversational systems, which now surpass traditional capabilities in understanding and exhibiting human-like reasoning. However, they inherently lack the ability to associate temporal relevance with memories; unlike humans, they typically assign uniform importance to all stored information. Additionally, as interactions between users and LLMs grow, the absence of built-in memory consolidation leads to responses that progressively lose contextual richness. To address this, we propose a study of a novel system that autonomously constructs a memory structure for the LLM, enabling holistic interpretations of users, situations, and environments. With each data point for an individual, the system extracts features to interpret and convert into a memory, which is subsequently assigned a rank based on a mathematical model incorporating contextual relevance, time elapsed, recall frequency, behavior, and engagement patterns. This ranking mechanism empowers the LLM to generate interpretations that

closely approximate human-like reasoning. With this architecture in place, personalized recommendations can be effectively delivered across domains such as healthcare and wellness, education and learning, social relationships, entertainment, and beyond.

CCS Concepts

• Ubiquitous Computing; • Large Language models; • Memory Consolidation;

Keywords

Memory Retrieval Models; Relevance; Recall Probability; Memory Ranking

ACM Reference Format:

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1 Introduction

Ubiquitous computing aims to embed computing into everyday life through pervasive technologies such as sensors, PDAs, and smart devices, enabling seamless data processing across environments without explicit user input [13]. As these systems generate continuous streams of context-aware data, they present new opportunities



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to enhance intelligent models—particularly large language models (LLMs).

Despite their capabilities, LLMs struggle with integrating long, continuous input. They often underperform on information positioned mid-sequence—a limitation known as the "lost in the middle" problem [10]. Additionally, longer sequences increase attention noise, diluting token relevance and reducing performance.

To mitigate these issues, we propose using ubiquitous computing as a complementary mechanism. Distributed sensor networks can capture fine-grained behavioral and contextual data, offering a more structured memory foundation for LLMs. This is especially valuable in domains like student subject selection, where early life decisions have long-term societal and economic impacts. Education, in particular, is a key driver of productivity and living standard improvements [12].

We used the En-Gage dataset from a K–12 field study in Melbourne [3], which includes multimodal sensor data capturing student engagement and behavior. Features such as stress, context, and physiological signals were extracted from classroom sessions. This data was converted into natural language and ranked by relevance for LLM consumption. Ranked entries were stored to enable selective, memory-relevant retrieval. This structured approach allowed LLMs to generate personalized subject recommendations based on individual engagement profiles.

Ultimately, this research proposes a novel memory consolidation framework that enhances LLMs' ability to assimilate and utilize pertinent, goal-specific information, advancing personalized educational guidance grounded in robust, context-rich data.

2 Related Work

This section reviews existing literature.

2.1 LLMs and Context-Aware IoT Systems in Ubiquitous Computing

LLM-driven Context-aware Computing (LCAc) uses natural language representations of sensor data and user queries through autonomous agents for context reasoning and action planning. Recent IoT-LLM frameworks [1] extend this by simplifying sensor data, augmenting IoT knowledge retrieval, and tailoring prompts for diverse IoT tasks.

2.2 Mathematical Models of Memory Consolidation and Recall

Neurocognitive models provide a basis for simulating memory consolidation in LLMs. Chessa and Murre's memory chain model [2] expresses recall probability through time-dependent neural activation, with memory strength decaying exponentially via an intensity function:

$$r(t) = \mu e^{-at},$$

where μ is the initial memory strength and a is the decay rate. The resulting recall probability for a single memory becomes:

$$p(t) = 1 - \exp(-\mu e^{-at})$$

reflecting the classic pattern of short-term memory decline, as seen in cognitive studies like the Brown-Peterson task [11].

This model captures recall probabilities using functions derived from non-homogeneous Poisson processes, underpinned by the assumption that neuronal firing occurs independently and stochastically over time [4].

To account for semantic relevance in LLM applications, Hou et al. [5] extend this framework by incorporating cosine similarity:

$$r = \frac{a \cdot b}{||a|| ||b||},$$

between content embeddings, thereby modulating the recall likelihood based on contextual similarity. Repeated recalls further influence memory strength by adapting the decay rate using a consolidation function:

$$a = \frac{1}{g(n)}, \quad g(0) = 1, \quad g(n) = g(n-1) + S(t),$$

The normalized recall probability after n recalls becomes:

$$P_n(t) = \frac{1 - \exp(-re^{-t/g(n)})}{1 - e^{-1}},$$

capturing the interplay of semantic relevance and recall frequency on retention.

These models guide the design of our memory consolidation mechanism for LLMs, enabling dynamic, relevance-driven retention and more cognitively plausible retrieval behavior.

3 Methodology

3.1 Model

We employed the mathematical memory consolidation framework from [5] to infer engagement levels from classroom sensor data collected via wearable devices [3]. Calculating recall probability was critical since sensor data is time-stamped, requiring accurate engagement and stress level deduction over class duration for effective memory consolidation within the LLM.

The core mathematical model we adopted is defined by:

$$P_n(t) = \frac{1 - \exp(-re^{-t/g(n)})}{1 - e^{-1}}$$

Relevance r is computed via sentence-transformer embeddings, with semantic similarity outperforming cosine similarity for behavioral signals. Time t models exponential decay between closely spaced sensor memories.

We introduced a consolidation factor g_n to adaptively strengthen memory traces based on recall frequency, slowing decay for recurring patterns while letting less relevant fluctuations fade.

The consolidation factor was updated with each new memory input according to:

$$g_n = g_{n-1} + \frac{1 - e^{-t}}{1 + e^{-t}}$$

Time t is normalized by temporal scaling (division by 100.0) ensuring typical sensor intervals map meaningfully to the modified sigmoid function $S(t)$. This prevents saturation at small time differences while preserving temporal sensitivity. The additive formulation $g_n = g_{n-1} + S(t)$ ensures each recall incrementally strengthens memory traces, with strengthening modulated by time since the last related event, creating time-sensitive consolidation aligned with natural cognitive processes.

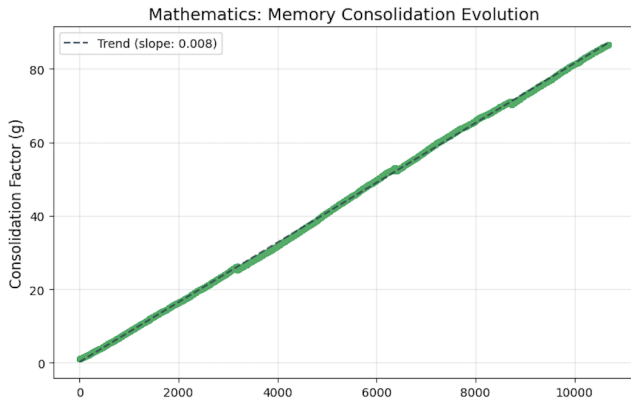


Figure 2: Memory consolidation factor g_n evolution over time, showing linear growth (slope = 0.008) from initial value 1 to maximum 86.563, demonstrating progressive strengthening of recalled sensor patterns.

4 Experimental Setup and Potential Results

4.1 Dataset

Experiments were conducted in Python using the In-Gauge and En-Gage datasets [3]. Ground truth from Classtable.csv (week, week-day, times, subject) aligned survey-based engagement scores, thermal sensation, and location labels with wearable sensor data via class IDs and temporal matching.

4.2 Data Processing Pipeline

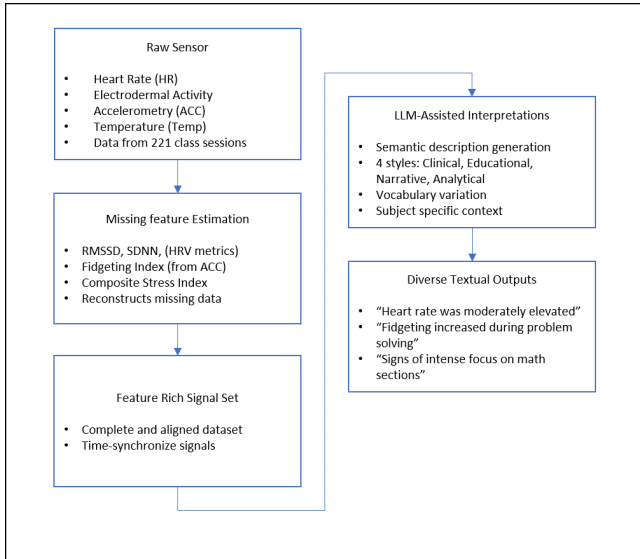


Figure 3: LLM-based pipeline for physiological interpretation. Raw multimodal signals are preprocessed and semantically interpreted into diverse textual forms using curated styles and contextual grounding.

We built a pipeline that converts raw physiological signals into semantically rich text for memory consolidation analysis (Fig. 3). Multimodal data from 221 classroom sessions undergo missing feature estimation (e.g., HRV, fidgeting index, stress score) to form time-aligned profiles, which are then processed by an LLM to generate clinical, educational, narrative, and analytical descriptions contextualized to classroom activities.

4.3 Memory Consolidation and Engagement Validation

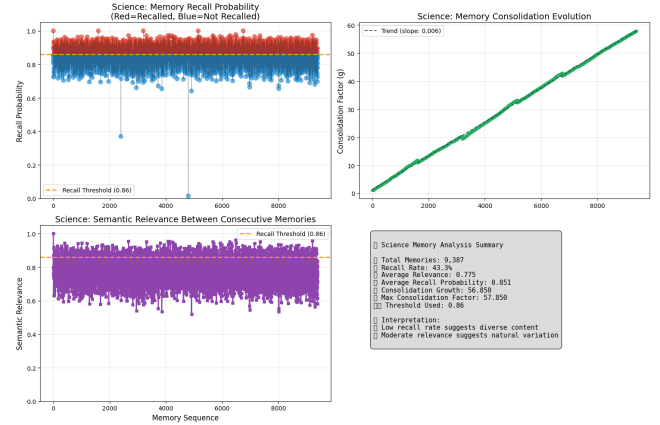


Figure 4: Average recall probability of 0.851 for threshold-exceeding memories in Science sessions, indicating robust memory consolidation patterns with a maximum consolidation factor of 57.850.

Day-wise Engagement Score Calculation and Validation: To model engagement through memory consolidation, we computed daily engagement scores by aggregating recalled memory traces that exceeded a semantic similarity threshold (0.86), along with their average consolidation factors and maximum recall probabilities. These scores were discretized into categorical levels (Low: 0–2, Moderate: 2–4, High: 4–5) to support interpretable comparisons.

As a baseline, we used the LLaSA model [6] in combination with the NousResearch/Llama-2-7b-chat-hf language model [8][7][9] to generate engagement-related interpretations directly from physiological sensor data, without incorporating any memory consolidation mechanism. We then evaluated our proposed model, which embeds a neurocognitively grounded memory structure and uses GPT-4o for interpretation over consolidated memory traces. This setup allowed the model to reason across temporally distributed, semantically filtered information, improving context-aware engagement inference.

To validate the models, we used ground truth engagement scores from 488 student responses in Studentsurvey.csv, computed by reverse-coding items 2 and 4 and averaging across five Likert-scale dimensions. Because engagement is highly individual, validation was performed separately for 10 students. Correlation analyses and confusion matrices indicated that our memory-augmented GPT-4o model aligned slightly better with student-reported engagement than the LLaSA + Llama-2 baseline, demonstrating the benefit of

Table 1: Summary of memory consolidation metrics by subject for participant ID = 5. This shows the distribution of total memories, recall rates, semantic relevance, and memory slope, highlighting distinct engagement profiles across academic content. PE: Physical Education

Subject	Total Memories	Recall Rate (%)	Avg Relevance	Slope
Assembly	1,509	69.2	0.815	0.00949
PE	4,797	58.4	0.798	0.00801
Language	12,630	54.1	0.791	0.00791
Science	9,884	50.6	0.785	0.00714
English	12,136	49.3	0.783	0.00726

structured memory integration for personalized engagement modeling.

5 Preliminary Results

The memory consolidation analysis for participant ID = 5 (Table 1) reveals subject-specific engagement patterns. Assembly shows the highest recall rate (69.2%) despite fewer memories, suggesting strong immediate attention during brief, collective sessions. Physical Education exhibits moderate recall (58.4%) but the highest consolidation slope, indicating sustained memory strengthening over time. Language and English, while generating the most memories, show lower immediate recall but higher long-term consolidation, reflecting gradual engagement through repeated exposure. Science falls in between with balanced metrics. Overall, shorter sessions appear to drive immediate engagement, while longer academic subjects benefit from cumulative consolidation.

Table 2: Average correlation and F1 scores for engagement prediction across 10 students.

Model	Correlation (r)	F1 Score
LLaSA + Llama-2 (baseline)	0.62	0.68
Our Model (Memory + GPT-4o)	0.70	0.73

Table 2 reports the average Pearson correlation and macro-averaged F1 score across 10 students, comparing our memory-augmented GPT-4o model against the baseline model using LLaSA with Llama-2. For each student, model-generated engagement scores were categorized into three bins—Low (0–2), Moderate (2–4), and High (4–5)—based on memory consolidation metrics. These were compared against ground-truth engagement levels derived from the *Studentsurvey.csv*, where responses were averaged over five Likert-scale items after reverse-coding items 2 and 4.

The correlation (r) captures alignment between memory-derived and self-reported engagement scores, while the F1 score evaluates classification consistency across three engagement categories. Improvements in both metrics show that memory consolidation enhances engagement modeling fidelity.

6 Conclusion

This research addresses a critical educational challenge: enabling personalized subject selection to reduce misaligned academic pathways. By deploying LLM-driven analyses of ranked physiological memories, our system provides evidence-based recommendations—such as suggesting advanced Mathematics for students with consistently strong consolidation in analytical contexts, or proposing hands-on vocational tracks for those showing heightened engagement during kinesthetic activities. Compared to the LLaSA + Llama-2 baseline, our GPT-4o model with structured memory demonstrated slightly better alignment with self-reported engagement scores. While current results are based on a single case study, future work should expand validation to a broader student population, enabling scalable, real-time guidance for aligning curricular choices with authentic cognitive engagement profiles.

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Author Contributions

Parampuneet Kaur Thind: Writing – review and editing, Validation, Supervision, Software, Resources, Project administration

Vaibhav Kumar Katturu: Writing – original draft, Conceptualization, Visualization, Data Curation, Investigation

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